SOFTWARE ENGINEERING

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Software Project Management

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Why do Project fail ?

Following are the reasons for project failure:

- Failure in **time** management.
- Failure in **cost** management
- Failure in **scope** management
- Failure in **quality** management

Who saves us from Project Failure?

It's a Project Manager!!

The key skills that are essential for a project manager are:

1. Technical Skills

- **Understanding Software Development Life Cycle (SDLC):** Knowledge of various SDLC models (Waterfall, Agile, etc.) to effectively manage projects.

- **Familiarity with Programming Concepts:** Basic understanding of programming languages and software architecture to communicate effectively with technical teams.

- **Tools Proficiency:** Proficiency in project management tools (e.g., JIRA, Trello, Microsoft Project) and software development tools (e.g., Git)

2. Project Management Skills

- **Planning and Scheduling:** Ability to create detailed project plans, define milestones, and establish timelines using techniques like Gantt charts or PERT diagrams.

- **Risk Management:** Identifying potential risks, assessing their impact, and developing mitigation strategies to minimize disruptions.

- **Budgeting and Cost Management:** Skills in estimating project costs, managing budgets, and ensuring financial accountability throughout the project lifecycle.

3. Leadership Skills

- **Team Leadership:** Ability to inspire and motivate team members, fostering collaboration and a positive work environment.
- **Conflict Resolution:** Skills in mediating disputes and resolving conflicts within the team or with stakeholders effectively.
- **Decision-Making:** Strong analytical skills to make informed decisions quickly, especially under pressure.

4. Communication Skills

- **Verbal and Written Communication:** Clear communication of project goals, progress updates, and technical concepts to both technical and non-technical stakeholders.
- **Stakeholder Management:** Engaging with stakeholders to understand their needs, expectations, and concerns throughout the project lifecycle.
- **Presentation Skills:** Ability to present project status reports and proposals effectively to stakeholders and senior management.

5. Quality Assurance

- **Understanding Quality Standards:** Familiarity with quality assurance practices and standards (e.g., ISO 9001) to ensure deliverables meet required specifications.

- **Continuous Improvement:** Promoting a culture of continuous improvement through retrospectives and feedback loops.

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- **A software project manager** is a person who undertakes the responsibility of executing the software project.

- Project manager may never directly involve in producing the end product but he **controls and manages** the activities involved in production.

- A project manager closely monitors the development process, prepares and executes various plans, arranges necessary and adequate resources, maintains communication among all team members in order to address issues of cost, budget, resources, time, quality and customer satisfaction.

Let us see few **responsibilities** on a project manager shoulders –

1. **Managing People**
2. **Managing Projects**

Managing People:

- Act as project leader
- Liaison (work together) with stakeholders
- Managing human resources
- Setting up reporting hierarchy etc.

Managing Project

- Defining and setting up project scope
- Managing project management activities
- Monitoring progress and performance
- Risk analysis at every phase
- Take necessary step to avoid or come out of problems
- Act as project spokesperson

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Project Management

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Project Management activities

Definition:

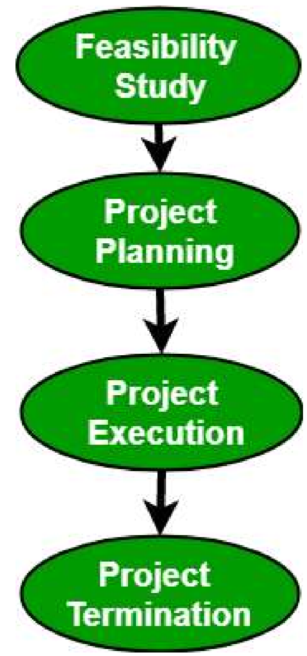
Project Management is an **art of managing all the aspects of a project** from inception to closure using a **scientific and structured** methodology.

The project must create something **unique** whether it is a **product**, **service** or **result** and must be progressively elaborated.

Project Management is the application of **knowledge**, **skills**, **tools** **and** **techniques** to project activities to meet the project requirements.

Project Management Process consists of the following 4 stages:

1. Feasibility study
2. Project Planning
3. Project Execution
4. Project Termination



1. Feasibility Study:

- **Feasibility** is defined as the practical extent to which a project can be performed successfully.

- To evaluate **feasibility**, a **feasibility study** is performed, which determines whether the solution considered to accomplish the requirements is practical and workable in the software.

- There are several fields of feasibility study including:

- **Economic feasibility,**
- **Operational feasibility,**
- **Technical feasibility.**

- The goal is to determine whether the system can be implemented or not.

2. Project Planning:

A detailed plan stating stepwise strategy to achieve the listed objectives is an integral part of any project.

Planning consists of the following activities:

- **Set objectives or goals**
- **Develop strategies**
- **Develop project policies**
- **Determine courses of action**
- **Making planning decisions**
- **Set procedures and rules for the project**
- **Develop a software project plan**
- **Prepare budget**
- **Conduct risk management**
- **Document software project plans**

3. Project Execution:

- A project is executed by choosing an appropriate software development lifecycle (SDLC) **model**.

- It includes a number of steps including **requirements analysis, design, coding and implementation, testing, delivery and maintenance**.

- There are a number of factors that need to be considered while doing so including the **size of the system**, **the nature of the project**, **time and budget constraints, domain requirements, etc.**

4. Project Termination:

- There can be **several reasons** for the termination of a project.

➢ Though expecting a project to **terminate after successful completion** is conventional, but at times, a project may **also terminate without completion**.

- Projects have to be closed down when the requirements are not fulfilled according to given time and cost constraints.

Some of the reasons for failure include.

Some of the reasons for failure include:

- Fast changing technology
- Project running out of time
- Organizational politics
- Too much change in customer requirements
- Project exceeding budget or funds

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Software Project Management (SPM)

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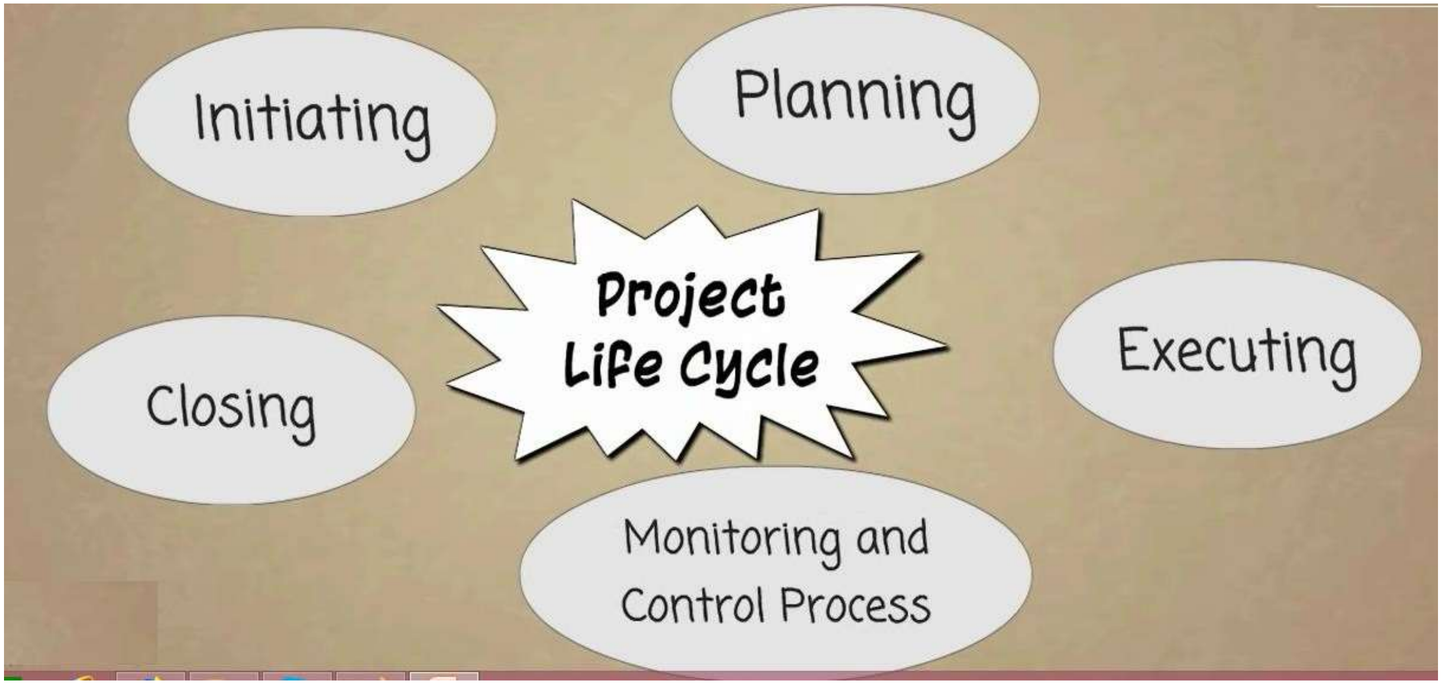
What is Software Project Management ?

- Software project management is the art and science of **planning** and **leading software projects.**

- It is a **sub-discipline** of project management in which software projects are **planned**, **implemented**, **monitored** and **controlled.**

- All projects are about **meeting objectives**.
- Like any other projects, a **software project** must **satisfy real needs**.
- To do this, **identify** the **project's stakeholders and their objectives.** Ensure that their **objectives are met** and this is the **aim of project management**

Phases of Software Project Management...



The Four important task in SPM are:

- Plan
- Organize
- Monitor
- Control

1. **Plan:** Planning is the foundation of software project management. It involves defining the project objectives, scope, deliverables, resources, timelines, and risks.
2. **Organize:** Organizing involves structuring the project team, assigning roles and responsibilities, and setting up processes and tools to execute the plan effectively.
3. **Monitor:** Monitoring involves tracking progress against the plan to ensure that the project stays on schedule, within budget, and meets quality standards.
4. **Control:** Control involves taking corrective actions based on monitoring results to keep the project aligned with its goals.

Effective software focuses on 3 Ps

People

Problem

Process

Effective software focuses on 3 Ps

People : To grow any business, you need the right people in place.

Problem : Has the product the brand sells been fine-tuned and optimized? Does it match the marketplace? Has a key target demographic been determined?

Process: A business must have a solid processes in place to thrive.

What is a Project ?

What is a Project? ...

- **Unique process**, consisting of a set of coordinated and controlled **activities** with **start and finish dates**, undertaken to achieve an **objective** conforming to specific requirements, including **constraints** of **time**, **cost** and **resources.**

- A project is an **activity** to meet the **creation** of a **unique product or service.**

- The activities that are undertaken to accomplish **routine activities cannot be considered as projects** *(but can be considered as jobs).*

Are Projects different from Operations ?

Projects Vs Operations (job)…

Projects	Operations
Temporary	On-going
Unique	Repetitive
Closed after attaining Objective	Objective is to sustain business

Project Vs Job…

- Jobs' – repetition of very well-defined and well understood tasks with very little uncertainty.

- 'Exploration' – the outcome is very uncertain

e.g. finding a cure for cancer (Some research projects)

- 'Projects' – in the middle

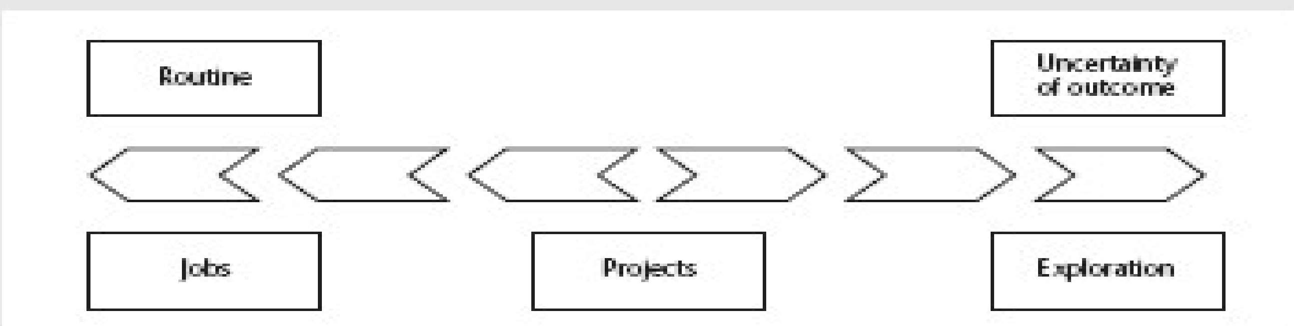


Figure 1.1 Activities most likely to benefit from project management

Characteristics of Project:

Following characteristics distinguish projects:

- Non-routine tasks are involved
- Planning is required
- Aiming at a specific target
- Predetermined time span
- Work carried out for a customer
- Work involves several specialism
- People are formed into a temporary work group to carry out the task
- Work is carried out in several different phases
- Constrained by time and resources
- Project is large and/or complex

Software Project Vs other types of project

1. **Invisibility**—when a physical artefact such as bridge is constructed the **progress can actually be seen** where as with software , progress is **not immediately visible**.

2. **Complexity**—per dollar, pound, euro spent, Software product contain more complexity than other engineered artefacts.

Software Project Vs other types of project…

3. Conformity -- the traditional engineer projects works with **physical systems** (materials like cement and steel). Whereas Software developers have to **conform to the requirements** of human clients.

4. Flexibility -- Software is **easy to change**, and is seen as a strength. Software will change to accommodate the other components rather than vice versa.

Projects can be:

- **In-house**: clients and developers are employed by the same organization.
- **Out-sourced:** clients and developers employed by different organizations.

- 'Project manager' could be:
  - a '**contract manager**' in the client organization.
  - a **technical project manager** in the supplier/services organization.

Activities covered by Software Project Management…

There are **3 successive processes/activities** that bring a new system.

1. Feasibility study

Is project technically feasible and worthwhile from a business point of view?

2. Planning

Only done if project is feasible.

For large project create **Outline plan** and detailed only for first stage.

3. Execution -

Often contains design and implementation sub-phases. Implement plan, but plan may be changed as we go along.

Activities covered by Software Project Management…

The **<u>3 successive processes/activities</u>** that bring a new system.

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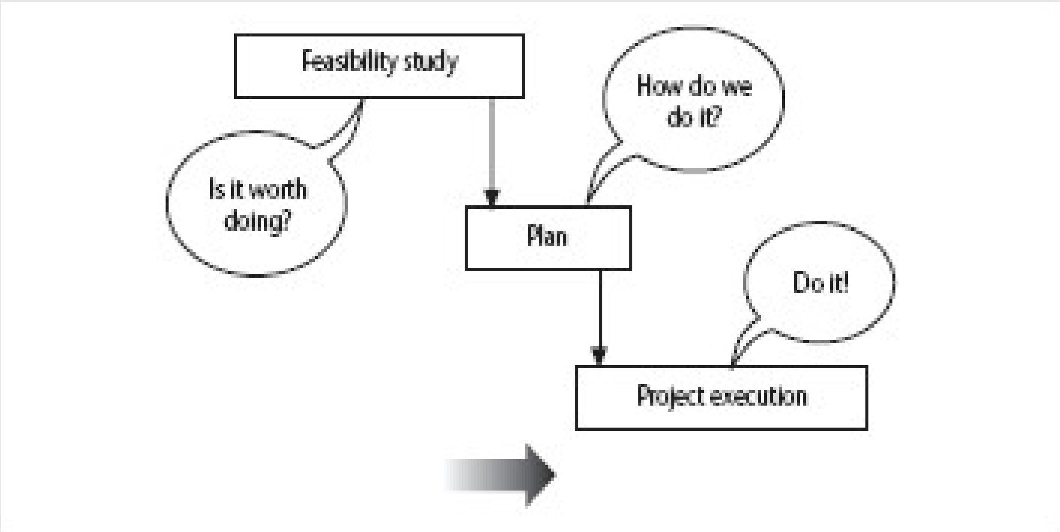

Figure 1.2 The feasibility study/plan/execution cycle

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Project Evaluation

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- **Project evaluation** is normally carried out **as first step of project planning**.

Project evaluation is a step by step process of **collecting, recording and organizing** information about

- **Project's result**
- **short - term outputs** (immediate results of activities or project deliverables)
- **Long – term outputs** (changes in behaviour , practice or policy resulting from the result)

The individual project **feasibility is evaluated** **Technical Assessment –**

Technical assessment of proposed system consists of evaluating:

- Whether the **required functionality** can be achie**ve**d with current affordable technologies.
- The cost of the technology adopted must be taken into account in the **cost-benefit analysis**.

Cost Benefit Analysis (CBA)

- Any project aiming at **return on investment** -- **must**, as a minimum, provide a greater benefit than putting that investment in bank.
- It is often necessary to **decide** if the proposed project is the best of several options.
- Not all projects can be undertaken.

- **Estimating the overall costs and benefits** of a project by producing a **cash flow forecasting** →indicates the **expenditure and income taken place** .
  - **A forecast is needed** of when expenditure, such as payment of salaries, and any income are to be expected.
  - There are some **methods for comparing projects** on the basis of their **cash flow forecasts**.

- It is assumed that the cash flows take place at the **end of each year.**
  - For short-term projects, there are **seasonal cash flow patterns**, some are **quarterly** or even monthly.

Following are some of the **cost-benefit evaluation techniques**:

1. Net profit

2. Payback period

3. Return On Investment (ROI)

4. Net Present Value (NPV)

5. Internal Rate of Return (IRR)

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Net Profit

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1. Net Profit

- The **Net profit** of a project is the difference between the **total costs** and the **total income** over the life of the project.
- I,e., Net profit is calculated by **subtracting a company's total expenses** from **total income**
- Net profit show what the company has earned (or lost) in a given period of time (usually one year)
- It is also called **net income** or **net earnings**

1. Net Profit

Net profit = total income - total cost

For Example: Consider the project cash flow estimates shown in the given table.

- Here negative values represent expenditure and positive values represent income

Example: Net Profit calculation

Year	Cash-flow
0	-100,000
1	10,000
2	10,000
3	10,000
4	20,000
5	100,000
Net profit	50,000

- Year-0→represents initial investment made at the start of the project.
- 'Cash-flow' is value of income less outgoing .
- Net profit value of all the cash-flows for the lifetime of the application

Activity: Net Profit calculation

- Calculate Net profit for the **project 2**

Year	Project-2 (Cash-Flow)
0	-1,000,000
1	200,000
2	200,000
3	200,000
4	200,000
5	300,000
Net Profit	

Activity: Net Profit calculation

- Calculate Net profit for the **project 3**

Year	Project-3 (Cash-Flow)
0	-100,000
1	30,000
2	30,000
3	30,000
4	30,000
5	30,000
Net Profit	

Activity: Net Profit calculation

- Calculate Net profit for the **project 4**

Year	Project-4 (Cash-Flow)
0	-120,000
1	30,000
2	30,000
3	30,000
4	30,000
5	75,000
Net Profit	

Activity: Net Profit calculation

Answer: Net profit for the project 2, 3 and 4 are as shown

Year	Project-2 (Cash-Flow)	Project-3 (Cash-Flow)	Project-4 (Cash-Flow)
0	-1,000,000	-100,000	-120,000
1	200,000	30,000	30,000
2	200,000	30,000	30,000
3	200,000	30,000	30,000
4	200,000	30,000	30,000
5	300,000	30,000	75,000
Net Profit	100,000	50,000	75,000

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Payback Period

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2. Payback Period

- The payback period is the time taken to break even or **pay back** the **initial investment**.

- Normally the project with the **shortest payback period will be chosen** on the basis that an organization will wish to **minimize the time that a project is 'in debt'**

Example: Payback Period Calculation

Year	Cash-flow (project 1)	Accumulated
0	-100,000	-100,000
1	10,000	-90,000
2	10,000	-80,000
3	10,000	-70,000
4	20,000	-50,000
5	100,000	50,000

Hence, for the given example
The Payback Period=5 years

- Payback period is the **time taken** to **pay back the initial investment**.
- This is the time it takes to start generating a surplus of income over outgoings

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Exercise: Payback Period Calculation

Consider the cash-flows of project 2, 3 & 4 and Calculate **Payback period** for each of them

Year	Project-2 (Cash-Flow)	Project-3 (Cash-Flow)	Project-4 (Cash-Flow)
0	-1,000,000	-100,000	-120,000
1	200,000	30,000	30,000
2	200,000	30,000	30,000
3	200,000	30,000	30,000
4	200,000	30,000	30,000
5	300,000	30,000	75,000
Payback period =	5 years	4 years	End of 4^{th} year

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Return on Investment (ROI)

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3. Return on Investment (ROI)

- **ROI** → It provides a way of **comparing** the net profitability to the investment required.

- A performance measure used to evaluate the efficiency of an investment or to compare the efficiency of a number of different investments.

- Return on investment (ROI) is a metric used to denote **how much profit has been generated from an investment that's been made.**

3. Return on Investment (ROI)

- Return on Investment is also known as the **Accounting Rate of Return (ARR),** provides a way of comparing the **net profitability** to the **investment** required.

- The formula used to calculate ROI is:

$$\text{ROI} = \frac{\text{average annual profit}}{\text{total investment}} * 100$$

Where,

$$\text{Average Annual Profit} = \frac{\text{Net Profit}}{\text{Payback period}}$$

Example: Return on Investment (ROI)

Calculating the ROI for project-1, the net profit is 50,000 and the total investment is 100,000.

$$\text{ROI} = \frac{\text{Average annual profit}}{\text{Total investment}} \times 100$$

$$\text{Average Annual Profit} = \frac{\text{Net Profit}}{\text{Payback period}}$$

- Average annual profit = $\frac{50,000}{5}$ = 10,000

ROI = $\frac{10,000}{100,000}$ X 100

ROI = 10%

Year	Cash-flow
0	-100,000
1	10,000
2	10,000
3	10,000
4	20,000
5	100,000
Net profit	50,000

Exercise1: Return on Investment (ROI)

- Calculate the ROI for each of the below given projects 2, 3 & 4 and decide which, on the basis of this criterion , is the most worthwhile.

Year	Project-2 (Cash-Flow)	Project-3 (Cash-Flow)	Project-4 (Cash-Flow)
0	-1,000,000	-100,000	-120,000
1	200,000	30,000	30,000
2	200,000	30,000	30,000
3	200,000	30,000	30,000
4	200,000	30,000	30,000
5	300,000	30,000	75,000
Net Profit			

Exercise2: Return on Investment (ROI)

- Calculate the ROI for each of the below given projects 1, 2, & 3 and decide which, on the basis of this criterion , is the most worthwhile.

	Investment	Netprofit
Project1	150000	50000
Project2	1,000000	1,00000
Project3	450000	40,000

- **The period of above project is 5 years.**

Exercise3: Return on Investment (ROI)

- There are two projects x and y. each project requires an investment of rs 20,000. you are required to rank these projects according to the pay back method from the following information.

- Year projectx projecty
- 1 1000 2000
- 2 2000 4000
- 3 4000 6000
- 4 5000 8000
- 5 8000

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Net Present Value (NPV)

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What is Net Present Value (NPV)?

"Net present value is **the present value of the cash flows at the required rate of return** of a project compared to its **initial investment**".

- In practical terms, it is a **method of calculating the return on investment (ROI), for a project or expenditure**. By looking at all of the money you expect to make from the investment and translating those returns into today's dollars (Rupees), you can decide whether the project is worthwhile.

What do companies typically use NPV for?

- When a manager needs to compare projects and decide which one to pursue, there are generally **three options** available:

1. **internal rate of return (IRR)**,
2. **payback method**, and
3. **net present value (NPV).**

- NPV is the tool of choice for most financial analysts.

- There are **two reasons** for that.

1. NPV considers the *time value of money*, **translating future cash flows into today's dollars.**

2. It provides a concrete number that managers can use to easily compare an initial outlay of cash against the present value of the return.

Net Present Value (NPV)

- Net Present value considers **profitability of a project** and the timing of each cash flows that are produced

$$\text{NPV} = \frac{\text{Sum of Expected Cash Flows For Each Period}}{(1 + \text{Discount Rate})^{\text{Period (year)}}} - \text{Initial Investment}$$

Or , **the above equation can be represented as follows**:

Key Formula:

$$NPV = \sum \left(\frac{CF_t}{(1+r)^t} \right) - \text{Initial Investment}$$

- CF_t = Cash flow in year t
- r = Discount rate
- t = Time period

- In order to **calculate NPV**, **we must** discount each **future cash flow** in order to get the **present value of each cash flow**, and then we **sum** those present values associated with each time period.

What does NPV equation determine?

- **Net present value** (**NPV**) **is** a method used to **determine** the current value of all future cash flows generated by a project, including the initial capital investment. It **is** widely used in capital budgeting to **establish** which projects **are** likely to turn the greatest profit.

- **Discounted Cash Flow (DCF):** is a cash flow summary adjusted to reflect the time value of money.
- DCF can be an important factor when evaluating or comparing investments, proposed actions, or purchases.
- Other things being equal, the action or investment with the larger DCF is the better decision.
- When discounted cash flow events in a cash flow stream are added together, the result is called the **Net Present Value (NPV).**

Net Present Value (NPV)

- When the analysis concerns a series of cash inflows or outflows coming at different future times, the series is called a **cash flow stream** .
- Each future cash flow has its own value today (its own **present value**).
- The **sum of these present values** is the **Net Present Value** for the cash flow stream.

- The size of the discounting effect depends on two things: the amount of time between now and each future payment (the number of discounting periods) and an interest rate called the **Discount Rate**

- As the number of discounting periods between now and the cash arrival increases, the present value decreases

$$\textbf{Discount factor} = \frac{1}{(1+r)^t}$$

Where , *r* is the interest rate(also called discount rate)

t is the number of years

For the given cash flow calculate the discount factor at the rate of 10%

Year	Project 1(Cash-flow)	Discount factor @ 10%
0	-100,000	
1	10,000	
2	10,000	
3	10,000	
4	20,000	
5	100,000	

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Net Present Value (NPV) - Solution

Year	Project 1(Cash-flow)	Discount factor @ 10%
0	-100,000	1.0000
1	10,000	0.9091
2	10,000	0.8264
3	10,000	0.7513
4	20,000	0.6830
5	100,000	0.6209

Net Present Value (NPV)

Year	Cash-flow	Discount factor(discount rate 10%)	Discounted cash flow
0	-100,000	1.0000	-100,000
1	10,000	0.9091	9,091
2	10,000	0.8264	8,264
3	10,000	0.7513	7,513
4	20,000	0.6830	13,660
5	100,000	0.6209	62,090
		NPV	618

Positive NPV:

- If present value of cash inflows is greater than the present value of the cash outflows, the net present value is said to be **positive** and the investment proposal is considered to be **acceptable**.

- If **NPV** is positive, that means that the **value** of the revenues (cash inflows) is greater than the costs (cash outflows). When revenues are greater than costs, the investor makes a profit. The opposite is true when the **NPV** is negative. When the **NPV** is 0, there is no gain or loss.

Zero NPV:

- If present value of cash inflow is equal to present value of cash outflow, the net present value is said to be **zero** and the investment proposal is considered to be **acceptable**.

For **example**, if a security offers a series of cash flows with an **NPV** of $50,000 and an investor pays exactly $50,000 for it, then the investor's **NPV** is $0.

Negative NPV:

- If present value of cash inflow is **less than** present value of cash outflow, the net present value is said to be **negative** and the investment proposal is **rejected**.

- **If NPV is negative** then it means that you're paying more than what the asset is worth. Zero **NPV**. **If NPV** is zero then it means you're paying exactly what the asset is worth.

Exercise 1: Net Present Value (NPV)

- Calculate NPV for the **project 2** **and discount factor of** **8%**

Year	Project-2 (Cash-Flow)
0	-1,000,000
1	200,000
2	200,000
3	200,000
4	200,000
5	300,000
Net Profit	

Exercise 2: Net Present Value (NPV)

- Calculate NPV for the **project 3** with a discount factor of **10%**

Year	Project-3 (Cash-Flow)
0	-100,000
1	30,000
2	30,000
3	30,000
4	30,000
5	30,000
Net Profit	

Exercise 3: Net Present Value (NPV)

- Calculate NPV for the **project 4** and discount factor **12%**

Year	Project-4 (Cash-Flow)
0	-120,000
1	30,000
2	30,000
3	30,000
4	30,000
5	75,000
Net Profit	

Summary point:

1. **NPV > 0** → **Profitable** (expected return > discount rate).
2. **NPV = 0** → **Break-even** (return = discount rate).
3. **NPV < 0** → **Unprofitable** (return < discount rate).
4. **Higher discount rates reduce NPV** (riskier projects need higher returns).

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Internal rate of return (IRR)

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Internal rate of return (IRR)

Definition:

- IRR is a **metric** **to estimate profitability of a project or business or investment**.
- It is a discount rate at which **NPV of project becomes "0"**

- In other words, **IRR is the annualized rate of return from an investment**

Formula

$$NPV = 0 = \sum \left(\frac{CF_t}{(1+IRR)^t} \right) - \text{Initial Investment}$$

CF_t = Cash flow in year t

IRR = Rate where NPV = 0

Decision Rule

- Accept if IRR > Required Rate of Return (e.g., cost of capital).
- Reject if IRR < Required Rate of Return.

IRR vs. NPV

- **NPV** gives absolute value ($), **IRR** gives percentage (%).
- **NPV is preferred** for conflicting results (IRR may mislead).

- **Note:**
- IRR is the **break-even return rate** for a project.
- Higher IRR = Better (if > hurdle rate).
- Use with **NPV** for robust decisions.

Hurdle Rate:

Definition

- The **hurdle rate** is the **minimum acceptable rate of return (MARR)** an investment must earn to be considered viable. It acts as a benchmark to decide whether to proceed with a project.

What is the purpose of hurdle rate?

It serve two purposes

- **Filters investments:** Only projects with returns **above** the hurdle rate are approved.
- **Reflects risk:** Higher-risk projects have higher hurdle rates.

Hurdle Rate vs. IRR

- **Hurdle Rate**: Predetermined minimum return (set by management).
- **IRR:** Actual return of the project (calculated from cash flows).

Formula:

$$\text{Hurdle Rate} = \text{Cost of Capital} + \text{Risk Premium}$$

Decision Rule

Accept if IRR > Hurdle Rate.

That is,

- A project should only be accepted if its IRR **is NOT less than** the hurdle rate, the **minimum** **required rate of return**.
  - The minimum required rate of return is based on the company's cost of capital and is adjusted to properly reflect the risk of the project.

- When comparing two or more mutually exclusive projects, the **project having highest value of IRR should be accepted**.

IRR Calculation

- There is no direct algebraic expression in which we might plug some numbers and get the IRR

- IRR is most commonly calculated using the **hit-and-trial** method.

- Since **IRR is defined as the discount rate at which NPV = 0**, we can write that:

  NPV = 0;

or

PV of future cash flows − Initial Investment = 0; or

$$\left[\frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \ldots\right] - \text{Initial Investment} = 0$$

Where,

r is the internal rate of return;

CF_1 is the period one net cash inflow;

CF_2 is the period two net cash inflow,

CF_3 is the period three net cash inflow, and so on…

The simplest method to use **hit and trial** as described below:

- **STEP 1:** Guess the value of r and calculate the NPV of the project at that value.
- **STEP 2:** If NPV is close to zero then IRR is equal to r.
- **STEP 3:** If NPV is greater than 0 then increase r and jump to step 5.
- **STEP 4:** If NPV is smaller than 0 then decrease r and jump to step 5.
- **STEP 5:** Recalculate NPV using the new value of r and go back to step 2.

Example 1: Basic IRR Calculation

Project: Invest $1,000 today. Get $500 after Year 1 and $800 after Year 2.

Solution:

1. **Set NPV = 0** and solve for IRR:

$$0 = -1000 + \frac{500}{(1+IRR)^1} + \frac{800}{(1+IRR)^2}$$

2. **Trial & Error:**
   - Try **IRR = 10%**:

$$NPV = -1000 + \frac{500}{1.10} + \frac{800}{1.21} = -1000 + 454.55 + 661.16 = +115.71$$

   - Try **IRR = 20%**:

$$NPV = -1000 + \frac{500}{1.20} + \frac{800}{1.44} = -1000 + 416.67 + 555.56 = -27.77$$

Exercise 1: Internal Rate of Return (IRR)

- Calculate **IRR** for the **project-2** at the discount factor of **12%**

Year	Project-2 (Cash-Flow)
0	-1,000,000
1	200,000
2	200,000
3	200,000
4	200,000
5	300,000
Net Profit	

Exercise 2: Internal Rate of Return (IRR)

- Calculate NPV & IRR for the **project-3** with a discount factor of 8%

Year	Project-3 (Cash-Flow)
0	-100,000
1	30,000
2	30,000
3	30,000
4	30,000
5	30,000
Net Profit	

Exercise 3: Internal Rate of Return (IRR)

- Calculate **NPV & IRR** for the **project 4** and discount factor **15%**

Year	Project-4 (Cash-Flow)
0	-120,000
1	30,000
2	30,000
3	30,000
4	30,000
5	75,000
Net Profit	

THANK YOU

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